

The Container Program

a reference overview

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Abstract

This is a navigable map of the container / directed-container / polynomial-functor program as it stands, not a publication. Each headline result is given in one to three sentences, tagged with its *honest* trust grade from the proof registry (`proofs/registry/*.json`) and its registry id so you can drill in. Grades run `[speculative] < [computed] < [proved] < [peer-reviewed] < [published] < [lean-verified]`. Nothing here is upgraded past its registry status; where a status is `[computed]` or `[in-progress]` I say so, and the degree-ladder of §2.6 is flagged throughout as a *program*, not a theorem.

Contents

1	Orientation	1
2	Theory	2
2.1	(a) The equivalence chain and the $\mathbf{DCont} \simeq \mathbf{Cat}$ hub	2
2.2	(b) Monoidal structures on containers / $\mathbf{Poly}$: the four tensors	2
2.3	(b') Day convolution	3
2.4	(c) Zappa–Szép / orchestration	3
2.5	(d) The tangent / Cartan-calculus arc	3
2.6	(e) The degree ladder / composition-blindness (a PROGRAM, not a theorem)	4
2.7	(f) The H^2 -cluster and the skew-brace obstruction	4
3	Applications	5
4	Formalisation (Lean 4)	5
5	State of the program (one-page table)	6

1 Orientation

The spine of the whole program is a single equivalence, read in four dialects:

$$\mathbf{Cont} \simeq \mathbf{DCont} \simeq \text{Polynomial comonoids in } (\mathbf{Poly}, \triangleleft) \simeq \mathbf{Cat} \text{ (small categories).}$$

Containers ($S \triangleleft P$) (Abbott–Altenkirch–Ghani) are shapes-and-positions; adding the Ahman–Chapman–Uustalu directed structure (D1–D5) makes the extension a comonad; a comonad for

substitution $\triangleleft$ is a comonoid in **Poly** (Spivak–Garner–Fairbanks); and such a comonoid is exactly a small category. **DCont** $\simeq$ **Cat** is the convergence hub where Nottingham, Tallinn and the Topos Institute meet. The central technical tool for *composition* is the Zappa–Szép product $K = C \bowtie D$ / distributive laws: it says when two directed containers (categories, update monads, orchestrations) can be welded into one, and the obstruction to welding is cohomological. Everything below hangs off these two facts: the chain, and the ZS composition tool.

2 Theory

2.1 (a) The equivalence chain and the **DCont** $\simeq$ **Cat** hub

- **The chain itself** is tracked as an *open umbrella node*, status [in-progress] (**equivalence-chain**). The classical equivalences (**Cont** $\simeq$ **DCont** $\simeq$ poly-comonoid $\simeq$ **Cat**) are each established and Lean-formalised as milestones M1–M4 (see §4); the research-level milestones M5 (bridge to **Poly**’s four tensors) and M6 (double category **Org**) remain open/conjectural. Treat “the equivalence chain” as a *scaffold with proved rungs*, not a single closed theorem.
- **DCont** $\cong$ **Cof**(**Poly**) (directed containers = cofree comonoids / small categories) is the hub result reused everywhere; the D1–D5 comonad laws are [lean-verified] (**cofree-comonad**, leaf **dcont-laws-D1–D5**).

2.2 (b) Monoidal structures on containers / **Poly**: the four tensors

Cont = **Fam**(**Set**^{op}) = **Poly** carries four monoidal products, each answering a different compositional question:

- ⊗ the **Dirichlet** / **Day convolution** tensor (parallel, symmetric) — see §2.3;
- $\triangleleft$ the **substitution** product (sequential composition of functors; its comonoids are the small categories);
- × the categorical product (independent choice);
- ⋈/⋈ the two **Dialectica** tensors (one-sided, non-convolutional).

Closedness / **closed structures**.

- (**Poly**, ×) is *cartesian closed*; **Poly** has **three** closed structures but is *not* locally cartesian closed. (**closed-day-structures**, [proved].)
- **T4 left $\triangleleft$ -closedness**: the non-symmetric $\triangleleft$ admits a *left* internal hom (distinct from the known $\triangleleft$ -coclosure = directed containers) over a rigid/tiny base, via the collapse $\triangleleft = \otimes$ there. [proved] (**t4-left-closedness-lhd-famcop**).
- **⋈/⋈ are the Dialectica tensors**: ⋈ restricts on the homogeneous subcategory to de Paiva’s Dialectica $\otimes$; they are non-convolutional, non-closed; non-closure is one-sided. Status [computed] (**other-cont-monoidal-tensors**).
- **Bare Dirichlet comonoids** = **Fam**(**Mon**^{op}): a $\otimes$ -comonoid in **Cont** with no $\triangleleft$ structure is exactly a family of monoids (cocommutative ones: **Fam**(**CMon**^{op})). [proved] and [lean-verified] **forward+converse** (**bare-dirichlet-comonoid**).

2.3 (b') Day convolution

The symmetric tensor on $\mathbf{Fam}(C^{\text{op}})$ is a Day convolution: for (S, P) and (T, Q) ,

$$(S, P) \otimes (T, Q) = (S \times T, (s, t) \mapsto P_s(x) Q_t),$$

convolving shapes by product and positions pointwise. Over $\mathbf{Set}$ this is Dirichlet multiplication of species; over a general closed symmetric monoidal base it is the Day convolution for the pointwise tensor of C .

- **T2 (Day-)closedness criterion** (Neil’s UID 124, “do T2 generally”): for a closed symmetric monoidal C with small coproducts, $(-) \otimes (T, Q)$ has a right adjoint (internal hom) inside $\mathbf{Fam}(C^{\text{op}})$ *iff* the functor $\Phi(Z) = \prod_{t \in T} \prod_{r \in R} C(M_r, Z(x)Q_t) : C \rightarrow \mathbf{Set}$ is *familiarily representable*. Over $\mathbf{Vec}$ this gives a clean dichotomy. [proved] (`t2-day-closedness-famcop`).
- The general **Day-convolutional tensor** $(\text{Cont}, \odot_*)$ is left closed *iff* $(-) \star B$ is polynomial for every set B , with a uniform Π -formula internal hom; this and “poly closed under composition” are [lean-verified] (`closed-day-structures`).
- The underlying full/faithfulness of container extension (the reason the Day picture is honest at all): $\langle\langle - \rangle\rangle : \mathbf{Fam}(C^{\text{op}}) \rightarrow [C, C]$ **is fully faithful iff the monoidal unit I is connected** (i.e. $C(I, -)$ preserves coproducts) — Neil’s flagship T1, *not* an extensivity condition. [proved] (`fullness-unit-connectedness`).

2.4 (c) Zappa–Szép / orchestration

- **Pairwise ZS criterion**: a pairwise-matched structure extends to a global Zappa–Szép product $K = C \bowtie D$ *iff* (L) freeness *and* (G) global closure hold. [proved] (`pairwise-zs`); the Lean ZS-associativity input is [lean-verified].
- **The (G) obstruction is an H^2 class**: $(G) \Leftrightarrow [\omega] = 0$. In the orchestration model, parametrising supervisor–worker by a token-mutation bit $\varepsilon \in \mathbb{Z}/2$, $C \bowtie D$ exists *iff* $\varepsilon = 0$ and $[\omega(K_\varepsilon)] = \varepsilon \cdot (\text{gen})$ in $H^2(\text{Sk}; \mathbb{Z}/2) \cong \mathbb{Z}/2$. An unprotected re-entrancy pattern *is* a (G) -failure. [proved] (`orchestration-zs`); the leaf $[\omega] = \varepsilon$ is [lean-verified].
- **Holonomy / composition bridge**: composing two update monads sharing a state set is a ZS product welding the state/isotropy-holonomy mode to the directed/re-entrancy mode. [peer-reviewed] (`holonomy-composition-zs-bridge`).

2.5 (d) The tangent / Cartan-calculus arc

- **∂ is a differential combinator**: the AAGM one-hole container derivative is the differential combinator of a Cartesian differential category $\mathbf{SPoly}$ of polynomial functors — equivalently the first-order tangent structure realised by dual-number substitution $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$. [proved] (`tangent-container-derivative`); arithmetic and the CDC bridge are [lean-verified].
- **Uniqueness of the representable tangent structure**: among representable first-order tangent structures, in the *corrected* ambient category — base change by the Weil algebra $W = \mathbb{N}[\varepsilon]/\varepsilon^2$, $T = (-) \otimes W$, on finite-rank Weil/ $\mathbb{N}$ -algebra objects — the only nontrivial one is ∂ . Status [proved], specifically *accept-with-required-revisions* from Rick’s 2026-10-07 referee report, which *supersedes* the earlier clean peer-review stamp; so this is **not** a clean peer-reviewed result (`tangent-uniqueness-spoly`). Note the ambient category is $T = (-) \otimes W$, *not* full substitution $(\text{Poly}, \triangleleft, y)$.

- **Negation obstruction:** on finite-support polynomial functors the additive bundle of $T = (-) \triangleleft D$ is a bundle of commutative *monoids* with no fiberwise inverses, so SPoly carries no scalar ring object and the full Cartan calculus $(d, \iota, \mathcal{L})$ does *not* descend — Cartan fails over $\mathbb{N}$. [proved], with [lean-verified] arithmetic (`cartan-scalar-object-spoly`).
- **The $\mathbb{Z}$ -completion far side:** completing the rig $\mathbb{N} \rightarrow \mathbb{Z}$ (virtual containers) switches Cartan back *on*. [proved] (`virtual-container-cartan`), $\mathbb{Z}$ -mirror [lean-verified].
- **Parallelizability does not port:** the attempt to port Lanfranchi’s “parallelizable $\Leftrightarrow$ Lie group object” is a *refutation with salvage* — the literal port is ill-posed ($T \not\lhd \times$); what survives is “tangent-regular $\Leftrightarrow$ out-degree homogeneous”, which is composition-*blind*. [proved] (`lie-parallelizable-poly-comonoid`), leaf `tangentRegular-outdegree` [lean-verified].

2.6 (e) The degree ladder / composition-blindness (a PROGRAM, not a theorem)

Remark. *The following is a **conjectural organising program**, not a proved theorem. The meta-claim “cohomological degree classifies compositional invariants” is a conjecture; the individual rungs below have their own (varying) grades. It is the “composition is a degree-two datum” synthesis, shipped 2026-10-07 as a program.*

The slogan: compositional invariants of a directed container stratify by cohomological degree.

- **Degree 0:** out-degree / tangent-regularity — composition-blind, sees only local fibre rank (`lie-parallelizable-poly-comonoid`, [proved]).
- **Degree 1:** supply-chain inventory consistency as a class $[\theta_R] \in H^1$, *independent* of the degree-2 weld (see §3, `supply-chain-zs`, [computed]).
- **Degree 2:** the ZS weld $[\omega] \in H^2$ (§2.4); plus the observation that δ -left-translation trivializability over a component sees exactly the *groupoids*, and the three-level monodromy tower (generic $\subsetneq$ groupoid $\subsetneq$ thin), separated by the vertex group. Both are registered as [proved]: “ δ -left-translation trivializable over a component $\Leftrightarrow$ groupoid” is node `delta-left-translation-trivialization` ([proved]), with crux child `arrows-out-presheaf-local-system-iff-groupoid` ([proved]); the constant-trivializable $\Leftrightarrow$ thin-groupoid refinement (the monodromy tower) is node `arrows-out-presheaf-const` ([proved]). Both live under the [proved] root `lie-parallelizable-poly-comonoid.json`.

2.7 (f) The H^2 -cluster and the skew-brace obstruction

Three genuinely different H^2 obstruction theories live on $\mathbf{DCont} \simeq \mathbf{Cat}$:

- **ZS holonomy** $[\omega]$ (§2.4, `orchestration-zs`).
- **Ferri prunability** — shown to be a degree-1 descent/subfunctor condition, orthogonal to $[\omega]$ (`memory-prunability-is-descent-not-schreier`; `ferri-prunability-separates-from-zs-holonomy`). The groupoid case: when the base is a connected groupoid the merge class is a Schreier extension class $H^2(\Gamma; M)$, which need *not* vanish — the seed “groupoid $\Rightarrow$ contractible $\Rightarrow H^2 = 0$ ” conjecture is **refuted**. Status [computed] (`groupoid-zs-obstruction`).
- $H^2_{\mathbf{CatB}}$ — the two $[\omega]$ sites are irreducibly distinct: the handoff class and the stabiliser class are both $\cong \mathbb{F}_2$ with generator ε but are *not* the same class (isotropy-restriction kills one). [proved] (`two-omega-sites`).

- **o_ν skew-brace solvability obstruction:** a group homomorphism $o_\nu : H_+^2 \rightarrow C^3/A(\mathbb{Z}_o^2)$ with $\text{im } \varphi^+ = \ker o_\nu$ — an additive class extends to a full skew-brace cocycle iff o_ν kills it. [proved] (solvability-obstruction-o-nu); the $|G| = 16$ lock and torsor slice are [lean-verified] (quiver-skew-brace-zs).

3 Applications

- **GA composition / diversity dynamics:** island-model migration topology determines diversity dynamics; categorical composition (ZS / distributive laws, the laxator as obstruction) is the invariant. Empirical (Langer–Vega GECCO 2026); the categorical model is the orchestration/ZS machinery above.
- **Supply-chain inventory consistency:** a supply chain is a directed container; composing two flows sharing a warehouse is a ZS product; inventory inconsistency between two routes is a class $[\omega] \in H^2(\text{Sk}; \mathbb{Z}/n) \cong \mathbb{Z}/n$ (measuring the discrepancy in *units*). The consistency datum is degree-1 $[\theta_R] \in H^1$ and is *independent* of the weld $[\omega] \in H^2$. Status [computed] (supply-chain-zs); H^1 arithmetic and the W_1/W_2 separation are [lean-verified].
- **Blockchain re-entrancy = failed distributive law:** a smart contract’s re-entrancy bug is a nonzero $[\omega] = \varepsilon$ in $H^2 \cong \mathbb{F}_2$ — the same (G)-failure as orchestration. [proved] / $[\omega] = \varepsilon$ [lean-verified] (orchestration-zs).
- **AI agent orchestration / tool-calling protocols as directed containers:** agents are containers (S = request types, $P(s)$ = responses); tool-calling protocols are directed containers; composing supervisors is a ZS weld. [proved] (orchestration-zs); write-up “Tool-calling protocols are directed containers” shipped 2026-09-19.
- **M-containers (monad-lifted):** $M\text{-Cont} = \text{Fam}(\text{Kl}(M)^{\text{op}})$; $\langle\langle - \rangle\rangle_M$ is always faithful, and full iff M is codense and every $(M-)^A$ is connected (affine $\Rightarrow$ connected). [proved] (m-containers); the composition criterion (M-containers compose iff M is polynomial) is [computed].

4 Formalisation (Lean 4)

Nodes whose registry *root* status is [lean-verified]: **free-monad-grafting** (free monad = $\triangleleft$ -monoid, three monoid laws as tree-grafting), **cyclic-h2-gcd-engine**, **cyclic-h2-norm-engine**, **fibre-h2-z2-engine** (the H^2 arithmetic/cohomology engines).

Lean-verified *leaves* inside otherwise-[proved] nodes (a representative list):

- Equivalence-chain milestones M1 representation, M2 comonad laws + converse, M3 comonoid forward+converse, M4 $\text{DCont} \cong \text{Cof}$, ZS, four-monoidal coherence (**equivalence-chain**).
- ZS associativity / $[\omega] = \varepsilon$ re-entrancy (**orchestration-zs, pairwise-zs**).
- Cofree comonad D1–D5 directed-container laws (**cofree-comonad**); free monad grafting laws (**free-monad-grafting**).
- $\text{Monad} \rightarrow \text{comonad}$ transfer, T_M cartesian boundary, Reader/State drop, bikleisli / effect-coeffect unit laws, affine monad=monoid, proof-relevance $\square$ (**monad-comonad-transfer, effect-coeffect-arrows**).

- Tangent: dual-number stage-1 arithmetic + finite tangent bridge (`tangent-container-derivative`); no-negation rig + fiber witnesses (`cartan-scalar-object-spoly`); solid-idempotent $n^2 = n +$ rank-rigidity + two-model distinction (`tangent-uniqueness-spoly`); $\mathbb{Z}$ -mirror (`virtual-container-cartan`); tangent-regular=out-degree (`lie-parallelizable-poly-comonoid`).
- Dirichlet/ $\times$ comonoid classification forward+converse (`bare-dirichlet-comonoid`, `dirichlet-monoid-classification`); Day II-form=morphism-form + poly closed under composition (`closed-day-structures`).
- Game-comonad (EF) comonoid witness (`game-comonad-poly`); QPF abs/repr collapse (memory `lean-qpf-collapse`).
- Skew-brace: o_ν torsor (+ UID200 converse), $|G| = 16$ lock, W_4 non-injective, additive-forgetful descends (`quiver-skew-brace-zs`, `solvability-obstruction-o-nu`).
- State-object δ / DCont $\simeq$ Cat hinge, worker composition (`state-object-delta`); workers-retract-of-BHM (`workers-retract-of-bhm-grading`); holonomy-triviality collapse (`holonomy-triviality`).

5 State of the program (one-page table)

Result	Pillar	Grade	Registry id
Equivalence chain (M1–M6 umbrella)	Theory (a)	in-progress	<code>equivalence-chain</code>
DCont $\cong$ Cof(Poly), D1–D5	Theory (a)	lean-verified (leaf)	<code>cofree-comonad</code>
Day/Dirichlet closedness (T2)	Theory (b')	proved	<code>t2-day-closedness-famcop</code>
Day-convolution left-closedness formula	Theory (b')	proved	<code>closed-day-structures</code>
Extension $f \& f \Leftrightarrow$ unit connected (T1)	Theory (b')	proved	<code>fullness-unit-connectedness</code>
T4 left $\triangleleft$ -closedness via collapse	Theory (b)	proved	<code>t4-left-closedness-lhd-famcop</code>
$\times / \rtimes$ are Dialectica tensors	Theory (b)	computed	<code>other-cont-monoidal-tensors</code>
Bare Dirichlet comonoid = Fam(Mon ^{op})	Theory (b)	proved	<code>bare-dirichlet-comonoid</code>
Pairwise ZS: $K = C \bowtie D \Leftrightarrow (L) \wedge (G)$	Theory (c)	proved	<code>pairwise-zs</code>
$(G) \Leftrightarrow [\omega] = 0$; orchestration dichotomy	Theory (c)	proved	<code>orchestration-zs</code>
Holonomy/composition ZS bridge	Theory (c)	peer-reviewed	<code>holonomy-composition-zs-bridge</code>
∂ = differential combinator (CDC)	Theory (d)	proved	<code>tangent-container-derivative</code>
Uniqueness of representable tangent ($T = (-) \otimes W$)	Theory (d)	proved (accept- w/-revisions)	<code>tangent-uniqueness-spoly</code>
Negation obstruction (Cartan fails over $\mathbb{N}$)	Theory (d)	proved	<code>cartan-scalar-object-spoly</code>
$\mathbb{Z}$ -completion switches Cartan on	Theory (d)	proved	<code>virtual-container-cartan</code>
Parallelizability port (refutation+salvage)	Theory (d)	proved	<code>lie-parallelizable-poly-comonoid</code>
Degree ladder META-claim (degree classifies composition)	Theory (e)	PROGRAM / conj.	(see §2.6)
deg-2 instance: δ -left-translation $\Leftrightarrow$ groupoid	Theory (e)	proved	<code>delta-left-translation-trivialization</code>
deg-2 instance: constant-triv. $\Leftrightarrow$ thin groupoid (monodromy tower)	Theory (e)	proved	<code>arrows-out-presheaf-constant-iff-thin</code>
Two ω -sites irreducibly distinct	Theory (f)	proved	<code>two-omega-sites</code>
Groupoid ZS obstruction (seed conj. refuted)	Theory (f)	computed	<code>groupoid-zs-obstruction</code>
o_ν solvability obstruction, $\text{im } \varphi^+ = \ker o_\nu$	Theory (f)	proved	<code>solvability-obstruction-o-nu</code>
Supply-chain inventory ($[\theta_R] \in H^1 \perp [\omega]$)	Applications	computed	<code>supply-chain-zs</code>
Blockchain re-entrancy $[\omega] = \varepsilon$	Applications	proved / lean-verified (leaf)	<code>orchestration-zs</code>
Agent orchestration / tool-calling protocols	Applications	proved	<code>orchestration-zs</code>

Result	Pillar	Grade	Registry id
M-containers = $\text{Fam}(\text{Kl}(M)^{\text{op}})$	Applications	proved	m-containers
Free monad = $\triangleleft$ -monoid (grafting laws)	Formalisation	lean-verified	free-monad-grafting
H^2 arithmetic/cohomology engines	Formalisation	lean-verified	cyclic-h2-*, fibre-h2-z2-engine
Game comonads factor through Poly	Theory/App	computed	game-comonad-poly

*Grades are as recorded in **proofs/registry/*.json** on 7 Oct 2026. The degree ladder is a program; “peer-reviewed” on tangent-uniqueness has been downgraded to accept-with-required-revisions by the 2026-10-07 referee report.*